SIMATS ENGINEERING

ASSESSMENT TOOL 2 - ANSWER KEY

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A - Scenario Based Answers

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Eng.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Number of Usable Host Addresses

- School of Computing (/25): 2^7 - 2 = 126 usable hosts
- School of Electronics (/26): 2^6 - 2 = 62 usable hosts
- School of Mechanical Engineering (/27): 2^5 - 2 = 30 usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Number of Usable Hosts

- Software Development (/24): 2^8 - 2 = 254 usable hosts
- Quality Assurance (/25): 2^7 - 2 = 126 usable hosts
- Human Resources (/26): 2^6 - 2 = 62 usable hosts

- Executive Management (/27): 2^5 - 2 = 30 usable hosts

4. Unused IP Address Range Remaining

After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits identify the network prefix:

Network Prefix = 2001:0db8:abcd:0000::/64

The remaining 64 bits (0000:0000:0000:0000) form the Interface ID, used to address individual hosts/devices.

4. Total Number of Host Addresses Supported in the Subnet

With a /64 prefix, 128 - 64 = 64 bits are available for host addressing.

Total addresses = 2^64 = 18,446,744,073,709,551,616 (approx. 1.8 x 10^19 addresses).

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does not reserve a separate network/broadcast address (as in IPv4), essentially all 2^64 addresses in the /64 subnet are usable for hosts/devices:

Total usable host addresses = 2^64 = 18,446,744,073,709,551,616, which is more than sufficient for thousands of medical devices and IoT sensors.

Answer 4: ISP Router - Longest Prefix / Routing Table Match

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4
- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

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After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

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After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

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3. Network Prefix (/64)

The first 64 bits identify the network prefix:

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Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

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(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

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Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

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3. Number of Usable Host Addresses

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4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

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Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

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Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

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Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
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Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

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(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

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Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

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3. Number of Usable Host Addresses

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4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

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Destination IP of incoming packet: 192.168.2.45

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Network Prefix = 2001:0db8:abcd:0000::/64

The remaining 64 bits (0000:0000:0000:0000) form the Interface ID, used to address individual hosts/devices.

4. Total Number of Host Addresses Supported in the Subnet

With a /64 prefix, 128 - 64 = 64 bits are available for host addressing.

Total addresses = 2^64 = 18,446,744,073,709,551,616 (approx. 1.8 x 10^19 addresses).

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does not reserve a separate network/broadcast address (as in IPv4), essentially all 2^64 addresses in the /64 subnet are usable for hosts/devices:

Total usable host addresses = 2^64 = 18,446,744,073,709,551,616, which is more than sufficient for thousands of medical devices and IoT sensors.

Answer 4: ISP Router - Longest Prefix / Routing Table Match

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4
- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Eng.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Number of Usable Host Addresses

- School of Computing (/25): 2^7 - 2 = 126 usable hosts
- School of Electronics (/26): 2^6 - 2 = 62 usable hosts
- School of Mechanical Engineering (/27): 2^5 - 2 = 30 usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Number of Usable Hosts

- Software Development (/24): 2^8 - 2 = 254 usable hosts
- Quality Assurance (/25): 2^7 - 2 = 126 usable hosts
- Human Resources (/26): 2^6 - 2 = 62 usable hosts
- Executive Management (/27): 2^5 - 2 = 30 usable hosts

4. Unused IP Address Range Remaining

After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits identify the network prefix:

Network Prefix = 2001:0db8:abcd:0000::/64

The remaining 64 bits (0000:0000:0000:0000) form the Interface ID, used to address individual hosts/devices.

4. Total Number of Host Addresses Supported in the Subnet

With a /64 prefix, 128 - 64 = 64 bits are available for host addressing.

Total addresses = 2^64 = 18,446,744,073,709,551,616 (approx. 1.8 x 10^19 addresses).

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does not reserve a separate network/broadcast address (as in IPv4), essentially all 2^64 addresses in the /64 subnet are usable for hosts/devices:

Total usable host addresses = 2^64 = 18,446,744,073,709,551,616, which is more than sufficient for thousands of medical devices and IoT sensors.

Answer 4: ISP Router - Longest Prefix / Routing Table Match

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4

- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Eng.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Number of Usable Host Addresses

- School of Computing (/25): 2^7 - 2 = 126 usable hosts
- School of Electronics (/26): 2^6 - 2 = 62 usable hosts
- School of Mechanical Engineering (/27): 2^5 - 2 = 30 usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Number of Usable Hosts

- Software Development (/24): 2^8 - 2 = 254 usable hosts
- Quality Assurance (/25): 2^7 - 2 = 126 usable hosts
- Human Resources (/26): 2^6 - 2 = 62 usable hosts
- Executive Management (/27): 2^5 - 2 = 30 usable hosts

4. Unused IP Address Range Remaining

After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits identify the network prefix:

Network Prefix = 2001:0db8:abcd:0000::/64

The remaining 64 bits (0000:0000:0000:0000) form the Interface ID, used to address individual hosts/devices.

4. Total Number of Host Addresses Supported in the Subnet

With a /64 prefix, 128 - 64 = 64 bits are available for host addressing.

Total addresses = 2^64 = 18,446,744,073,709,551,616 (approx. 1.8 x 10^19 addresses).

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does not reserve a separate network/broadcast address (as in IPv4), essentially all 2^64 addresses in the /64 subnet are usable for hosts/devices:

Total usable host addresses = 2^64 = 18,446,744,073,709,551,616, which is more than sufficient for thousands of medical devices and IoT sensors.

Answer 4: ISP Router - Longest Prefix / Routing Table Match

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4
- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Eng.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Number of Usable Host Addresses

- School of Computing (/25): $2^7 - 2 = 126$ usable hosts
- School of Electronics (/26): $2^6 - 2 = 62$ usable hosts
- School of Mechanical Engineering (/27): $2^5 - 2 = 30$ usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Number of Usable Hosts

- Software Development (/24): 2^8 - 2 = 254 usable hosts
- Quality Assurance (/25): 2^7 - 2 = 126 usable hosts
- Human Resources (/26): 2^6 - 2 = 62 usable hosts
- Executive Management (/27): 2^5 - 2 = 30 usable hosts

4. Unused IP Address Range Remaining

After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits identify the network prefix:

Network Prefix = 2001:0db8:abcd:0000::/64

The remaining 64 bits (0000:0000:0000:0000) form the Interface ID, used to address individual hosts/devices.

4. Total Number of Host Addresses Supported in the Subnet

With a /64 prefix, 128 - 64 = 64 bits are available for host addressing.

Total addresses = 2^64 = 18,446,744,073,709,551,616 (approx. 1.8 x 10^19 addresses).

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does not reserve a separate network/broadcast address (as in IPv4), essentially all 2^64 addresses in the /64 subnet are usable for hosts/devices:

Total usable host addresses = 2^64 = 18,446,744,073,709,551,616, which is more than sufficient for thousands of medical devices and IoT sensors.

Answer 4: ISP Router - Longest Prefix / Routing Table Match

Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

Administration LAN: 192.168.10.0/26, Gateway 192.168.10.1

Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4
- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65

Answer 1: Smart Campus Subnet Allocation (192.168.100.0/24)

Given block: 192.168.100.0/24, allocated as School of Computing (/25), School of Electronics (/26), School of Mechanical Engineering (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190
School of Mechanical Eng.	/27 (255.255.255.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222

3. Number of Usable Host Addresses

- School of Computing (/25): 2^7 - 2 = 126 usable hosts
- School of Electronics (/26): 2^6 - 2 = 62 usable hosts
- School of Mechanical Engineering (/27): 2^5 - 2 = 30 usable hosts

4. Remaining Unused IP Address Range

After allocating the three subnets, the remaining unused block within 192.168.100.0/24 is:

- 192.168.100.224 - 192.168.100.255 (a /27 block, 32 addresses, available for future use)

Answer 2: Multinational Company Subnet Allocation (10.10.0.0/16)

Given block: 10.10.0.0/16, allocated as Software Development (/24), Quality Assurance (/25), Human Resources (/26), Executive Management (/27).

1 & 2. Network Address, Broadcast Address and Valid Host Range

Department	Subnet Mask	Network Address	Broadcast Address	Valid Host Range
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Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222

3. Number of Usable Hosts

- Software Development (/24): 2^8 - 2 = 254 usable hosts
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4. Unused IP Address Range Remaining

After allocating the four department subnets, the unused range within the 10.10.0.0/16 block is:

- 10.10.1.224 - 10.10.255.255 (the remainder of the /16 block, available for future allocation)

Answer 3: Smart Hospital IPv6 Migration (2001:0db8:abcd::/64)

1. Compressed Notation

2001:db8:abcd::/64

(Leading zeros in each group are removed, and the contiguous group of all-zero fields is replaced with '::'.)

2. Expanded (Full Hexadecimal) Representation

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

The first 64 bits identify the network prefix:

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Routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

Destination IP of incoming packet: 192.168.2.45

1. Destination Subnet

192.168.2.45 falls within the range 192.168.2.0 - 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry

The router matches the entry 192.168.2.0/24, since the destination address lies within this subnet's range.

3. Next-Hop Network / Interface

The packet is forwarded out of the router interface that is directly connected to the 192.168.2.0/24 network (the interface configured with an IP address in that subnet).

4. Default Route (If No Match Found)

If the destination address does not match any specific entry in the routing table, the router forwards the packet using the default route:

Default Route = 0.0.0.0/0 (via the configured default gateway interface).

Answer 5: Two-Department LAN via Layer-3 Router

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Finance LAN: 192.168.10.64/26, Gateway 192.168.10.65

1 & 2. Network Address, Broadcast Address and Valid Host Range

LAN	Network Address	Broadcast Address	Valid Host Range
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Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126

3. Suggested IP Addresses for Three Computers in Each LAN

- Administration LAN: PC1 - 192.168.10.2, PC2 - 192.168.10.3, PC3 - 192.168.10.4
- Finance LAN: PC1 - 192.168.10.66, PC2 - 192.168.10.67, PC3 - 192.168.10.68

(Note: 192.168.10.1 and 192.168.10.65 are reserved for the router gateway interfaces and should not be assigned to hosts.)

4. Default Gateway for Each LAN

- Hosts in the Administration LAN use gateway: 192.168.10.1
- Hosts in the Finance LAN use gateway: 192.168.10.65